



HARDWARE-TRIGGERED METROLOGY TOPOLOGY

Escaping Software Timestamp Delusions via Logic Analyzer GPIO Toggles and Shunt Current Probes

Physical AI System Under Test (SUT)

Arduino UNO Q Dual-Brain Platform

1. Sensor Frame Interrupt (GPIO 1):

Toggled inside camera driver ISR on DMA start.

2. MPU Inference Emit (GPIO 2):

Toggled in Linux kernel upon writing RPMSG mailbox.

3. MCU Enforcement Veto / Pass (GPIO 3):

Toggled inside FreeRTOS 1 kHz ISR on CBF evaluation.

4. Inverter Gate Drive Out (PWM Pins):

Direct hardware timer registers driving MOSFET bridge.

Probe 1: Ingest

Probe 2: Inference

Probe 3: Enforce

Probe 4: Coil Current

External Ground-Truth Metrology

Multi-Channel Logic Analyzer & Oscilloscope

CH 1 (Digital): Transduction Pulse → True $t_{\text{transduce}}$

CH 2 (Digital): Proposal Generation → True $t_{\text{inference}}$

CH 3 (Digital): MCU Permission → True t_{enforce}

CH 4 (Analog): Current Shunt → Motor Coil L/R Rise Time

Guaranteed Result: Zero Operating System Jitter.